El Agente Creativo: Agentic Generation of Research Ideas with Representation Networks

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Abstract

We disentangle the contributions of explicit literature structure from those of the agentic loop in LLM research ideation. **El Agente Creativo** composes ideas by traversing a representation network distilled from source papers; a matched **Abstracts agent** baseline shares its LLM, workflow, output schema, and tool-call budget but reasons over the flat abstract corpus. Using an LLM-as-judge Elo protocol calibrated on ICLR oral-vs-poster gaps, we find that both agentic arms beat a deliberately strong shuffled LLM-only baseline by 57–123 Elo and exceed an ICLR 2026 oral anchor, yet are statistically indistinguishable from each other on quality within tournament noise—the headline gain is carried by the agentic loop, not by graph structure per se. Graph structure instead delivers a consistent nearest-neighbor diversity edge, largest cross-domain, at 1.29–$1.79\times$ lower per-idea *generation* token cost (a generation-time figure that excludes the graph's one-time construction, so the net cost advantage accrues only once that upfront build is amortized over enough ideas). Schema-isolating ablations trace this edge to the typed Repr/Transform schema: ReprNet beats a matched generic knowledge-graph agent by $+28.76$ to $+39.34$ Elo at matched cost, and removing the realization (is-a) hierarchy erases most of the gap. Agentic ideation is the right substrate; a typed schema with an abstraction hierarchy is a consistent diversity-and-cost win on top of it, most impactful when ideas must bridge unfamiliar domains.

1 Introduction

Large language models (LLMs) are increasingly capable partners in the scientific workflow, generating idea-like abstracts that read fluently and recover surface markers of novelty [Si et al., 2024, Kumar et al., 2025, Lu et al., 2024, Vaswani et al., 2023, Brown et al., 2020]. This progress, however, has been measured almost entirely by whether the output looks plausible, leaving open the question that matters for the design of practical scientific assistants: given a fixed set of source papers, does providing the LLM with a structured representation of the literature actually produce better ideas than letting the same model read the same abstracts directly?

This counterfactual is hard to answer because the obvious comparison is biased against the LLM-only baseline. If the LLM-only arm always sees source abstracts in the same order, it can be penalized by positional artifacts rather than by representation; if the two arms see different source pools, quality differences can be attributed to the pool rather than to structure. A genuine test of explicit structure therefore requires identical source identities, controlled abstract order, and a quality judge whose scale is itself externally validated—three constraints that have not been combined in prior ideation benchmarks.

We address this gap with **El Agente Creativo**, a graph-conditioned ideation agent that operates over a representation network in which each source paper is distilled into a small set of reusable *representation* nodes and *transform* nodes joined by typed edges, and that composes new ideas by bridging these nodes through retrieval and traversal tools. The agent is benchmarked against a deliberately strengthened LLM-only baseline: the same LLM, given the *same* source abstracts, with abstract order deterministically reshuffled per generated idea so that no fixed positional ordering can favor either arm. Across both within-domain (random ICLR) and cross-domain (Nature+ICLR) settings, this controls the counterfactual at the level of source identity, abstract ordering, and generation budget.

We pair this design with a calibration step that grounds the LLM-as-judge evaluator in a known-quality signal. Because LLM-as-judge tournaments are sensitive to prompt and judge bias, we precede every generated-idea claim with an oral-vs-poster calibration drawn from ICLR 2025 and ICLR 2026, recovering 66.0 and 35.1 Elo gaps that anchor an interpretive reference scale. All idea-quality comparisons in the paper are reported against this scale rather than as standalone Elo deltas, so the magnitude of the effect can be interpreted directly.

The study makes four contributions. **(i) Matched-source benchmarks.** We construct random10 and random20 ICLR 2025 benchmarks and an interdisciplinary Nature+ICLR benchmark in which LLM-only and ReprNet agents generate ideas from identical source-paper IDs and deterministically reshuffled abstract orders. **(ii) Calibrated quality evaluation.** We calibrate the Elo judge on oral-vs-poster distinctions, recovering 66.0 and 35.1 Elo gaps, and use these as an interpretive reference for generated-idea comparisons. **(iii) Consistent quality and diversity gains.** Graph-conditioned ideation improves mean Elo by 111.7–151.0 points across all three benchmarks, increases nearest-neighbor diversity in every setting, and exceeds the ICLR 2026 oral anchor in two of three tournaments. **(iv) End-to-end token observability.** We instrument every generation, traversal, and evaluation call, revealing that the ReprNet agent spends 43–95× more tokens per idea, concentrated in input-side traversal context rather than longer outputs, and identifying the most impactful direction for follow-on engineering.

2 Related Work

LLM-based research ideation. Prior ideation benchmarks have measured what LLMs can produce but not whether literature representation matters. Recent studies show that LLM-generated ideas can rival human-written proposals on novelty while still trailing on feasibility, grounding, and diversity [Si et al., 2024, Kumar et al., 2025]; IdeaBench supplies target-paper and reference-paper contexts for biomedical ideation, and AI Idea Bench scales the protocol to AI papers across multiple evaluation views [Guo et al., 2025, Qiu et al., 2025]. These efforts motivate our matched-source protocol, but none isolates the causal effect of the underlying literature representation. El Agente Creativo is designed specifically to ask the narrower counterfactual question: does a graph view of the same papers improve generation over flat prompting?

Structured representations for ideation. A growing literature already pre-structures the literature before ideation, but none of these systems isolates the representation effect. SciMON optimizes scientific inspirations for novelty, SciDeator recombines research-paper facets, and Idea-Catalyst decomposes research problems before retrieving and ranking cross-domain source insights [Wang et al., 2024, Radensky et al., 2024, Kargupta et al., 2026]. In parallel, knowledge-grounded and agentic discovery systems—KG-CoI, SciAgents, The AI Scientist, and Robin—wire graphs, tool use, or multi-agent loops into broader hypothesis-generation workflows [Xiong et al., 2024, Ghafarollahi and Buehler, 2024, Lu et al., 2024, Ghareeb et al., 2025]. El Agente Creativo is complementary to these systems: rather than building a full discovery agent, it isolates the representation layer and benchmarks graph-conditioned ideation head-to-head against a shuffled-abstract baseline that holds the source pool identical.

Evaluation methodology. Evaluation choices materially shape what one can claim about generated science, and existing protocols do not provide a calibrated scale. OpenNovelty grounds novelty assessment in retrieved supporting evidence, and ScholarEval targets literature-grounded research-idea evaluation [Zhang et al., 2026, Moussa et al., 2025]. Our protocol is lighter-weight than a full novelty audit but adds two features tailored to the present question: a calibration step that recovers the ICLR oral-vs-poster gap before any generated-idea claim is made, and per-call token observability that exposes the quality–cost tradeoff hidden inside agent pipelines.

3 The El Agente Creativo Pipeline

El Agente Creativo decouples scientific ideation into two stages so that explicit representation can be evaluated as an isolated design choice. In *graph construction*, each source paper is distilled into a compact set of representation nodes, transform nodes, and typed edges, with an extraction prompt that is budget-constrained so that each paper contributes its core contribution as a reusable abstraction rather than as verbatim transcription. In *graph-conditioned ideation*, an LLM agent traverses the resulting network through retrieval and traversal tools—sampling random nodes, expanding parent and child hierarchies, finding paths between distant concepts—and returns a single paper-style idea per run.

The LLM-only baseline matches the source papers, the deterministically reshuffled abstract order, the underlying LLM, and the decoding settings of the ReprNet agent, and returns an idea under the same JSON schema. Two implementation details differ beyond the representation, however, and we disclose them so the LLM-only gap is not over-read: the baseline is a single chat completion with no tool-use loop and a fixed output-token cap, and it recovers the idea by parsing free-form text rather than through the agents' structured-output validator with retries. The agentic-vs-LLM-only gap therefore conflates the agentic loop with output-format enforcement and the absence of an output cap. For this reason we treat LLM-only as a deliberately strong *lower bound* rather than a clean single-variable control, and rest the clean representation-only contrast on the ReprNet-vs-Abstracts comparison (§6), where the framework, schema, output handling, and budgets are matched and the substrate is the sole systematic difference.

A central design invariant is that prior experiment graphs are treated as read-only artifacts so that quality gains cannot be confounded by silent graph mutation. Every reuse of a cached graph proceeds through an experiment-scoped duplicate whose contents are verified against the source by SHA-256 fingerprints over node and edge payloads; the duplication and verification mechanics are described in Appendix A (Figure 1).

4 Experimental Design

Judge calibration. Every generated-idea claim in this paper is preceded by an oral-vs-poster calibration so that the Elo scale is externally validated before it is used to compare arms. The same Elo protocol is run over 100 ICLR 2025 oral abstracts versus 100 poster abstracts, and independently over the corresponding ICLR 2026 split, yielding two reference gaps against which all subsequent quality differences are read.

Matched random ICLR benchmark. The primary benchmark holds source identity constant across arms while sampling without submission-side filtering bias. Its source pool consists of 600 accepted ICLR 2025 papers obtained from OpenReview by querying accepted-venue counts with random offsets, partitioned into twenty disjoint random10 groups (pri-

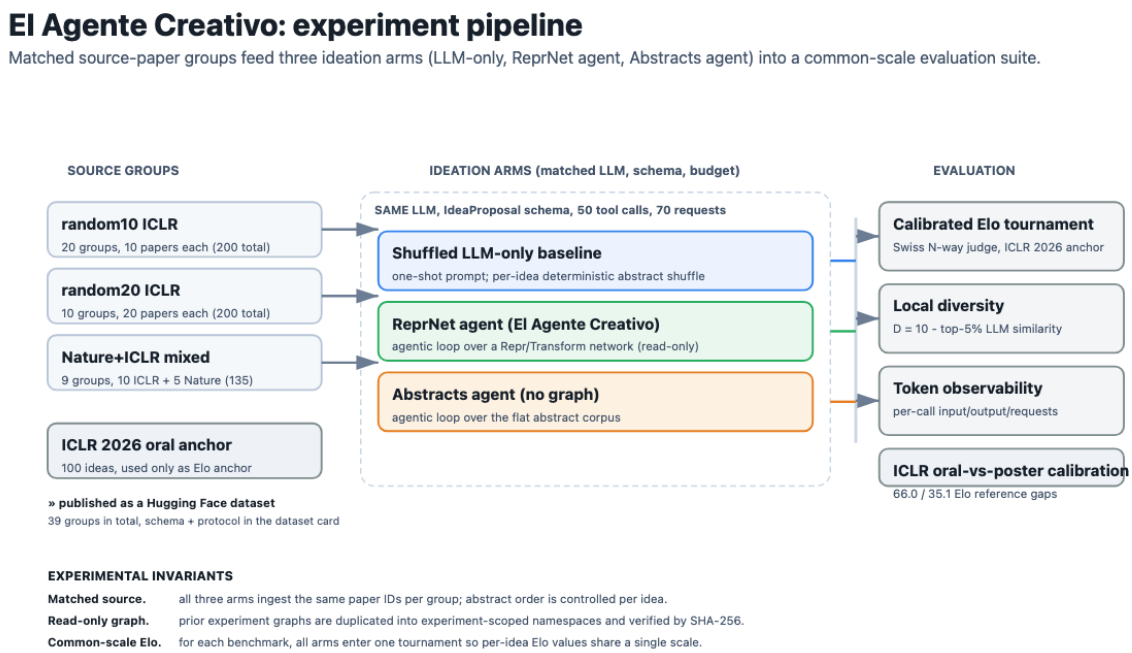

Figure 1: End-to-end experiment pipeline. Source paper abstracts are split into matched groups; the shuffled LLM-only baseline (top), the ReprNet agent (middle, reusing read-only representation networks from a prior run), and the Abstracts agent baseline (right of middle) each generate ideas from the *same* groups under matched LLM, schema, and tool-call budgets; outputs feed a unified Elo tournament, a nearest-neighbor diversity evaluator, and a per-call token-observability log.

mary condition) and ten disjoint random20 groups (secondary condition); each group produces ten LLM-only ideas and ten ReprNet-agent ideas. Abstract order is shuffled per idea with a deterministic seed so the LLM-only arm cannot be penalized by a fixed positional artifact, and the ReprNet agent reuses cached graphs through a fingerprint-verified read-only duplicate (Appendix A).

Abstracts agent (non-graph agentic baseline). To distinguish the contribution of explicit graph structure from that of the agentic loop, every benchmark in this paper also includes a third arm called **Abstracts agent**. Abstracts agent matches the ReprNet agent on the confounding degrees of freedom (same LLM, same Pydantic AI framework, same `IdeaProposal` schema, same source IDs, same 50 tool-call and 70 request budget, same workflow of sample $\rightarrow$ inspect $\rightarrow$ explore neighbors $\rightarrow$ bridge $\rightarrow$ synthesize $\rightarrow$ self-critique) but its tool palette (`random_paper`, `read_paper`, `search_papers`, `find_related`, `bridge`) acts on the flat abstract corpus rather than on the representation graph, using sentence-embedding cosine similarity in place of typed-edge traversal. The palettes are matched in workflow role rather than in exact tool count—the ReprNet read-only palette is somewhat larger (it adds hierarchy- and path-navigation tools that presuppose the graph substrate; Appendix A)—so the contrast isolates the substrate while, if anything, granting the graph arm a few extra affordances. Abstracts agent generates 10 ideas per group at each benchmark—200 at random10, 100 at random20, 90 at Nature+ICLR mixed—and enters the tournament for each setting alongside the LLM-only, ReprNet-agent, and (where applicable) ICLR 2026 oral anchor pools. Half-budget Elo tournaments (1200 matches for random10, 2400 for random20, 1800 for mixed) are used because the addition of a fourth arm doubles compute relative to the original two-arm V2 measurements; this leaves rms convergence below 10 Elo in all cases, sufficient to read the ReprNet-vs-Abstracts comparisons. The interdisciplinary methodology for the mixed setting mirrors the ReprNet agent's mixed methodology so both agentic arms face the same cross-domain constraints.

Interdisciplinary Nature+ICLR benchmark. A second benchmark tests whether the effect transfers to a genuinely cross-disciplinary setting where domain integration is the central challenge. It uses nine mixed groups, each starting from an ICLR random10 graph and augmented with five recent Nature-family abstracts spanning Nature Physics, Nature Biotechnology, and Nature Chemistry; LLM-only and ReprNet agents each generate ten ideas per group. Beyond Elo and diversity, a dedicated rubric scores each idea on five interdisciplinarity axes—domain coverage, integration depth, bidirectional value, source grounding, and feasibility—on a 1–5 scale.

5 Evaluation Methodology

5.1 Quality via Calibrated Elo

Idea quality is scored by a blinded LLM-as-judge tournament whose pairwise preferences drive a Bradley–Terry update over per-idea Elo ratings [Elo, 1978, Bradley and Terry, 1952]. All ideas in a setting begin at Elo 1000; each match presents $m = 4$ ideas chosen by a Swiss-style scheduler that balances participant exposure, prefers nearby Elo ratings after a warm-up phase, and retains an exploration term so mid-rank pairings remain well sampled [Zheng et al., 2023, Chiang et al., 2024]; presentation order is shuffled before judging. The judge returns a complete ranking which is decomposed into all $m(m-1)/2$ pairwise comparisons and applied via the standard Bradley–Terry expected-score update with $K = 32$ (see Appendix A for the exact equations). The reported quality score for each arm is its mean final Elo, with nonparametric bootstrap 95% confidence intervals over idea-level Elo values. Because the same protocol is first applied to oral-vs-poster calibration sets, every generated-idea gap is read against a known-quality reference scale rather than as an isolated number.

5.2 Local Diversity

A second axis is required because Elo can reward a base of near-duplicate ideas without penalty. We therefore measure *local self-similarity* inside each idea base: for every idea, we embed its title and abstract, retrieve the top 5% nearest neighbors by cosine similarity, deduplicate pairs, and cap the evaluation at the configured budget, after which an LLM rates each pair on a 1–10 semantic-similarity scale. Following the spirit of Self-BLEU-style redundancy checks [Zhu et al., 2018], we report

$$D = 10 - \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} s(i,j), \tag{1}$$

where $\mathcal{P}$ is the selected nearest-neighbor pair set and $s(i,j)$ the LLM similarity score; higher D corresponds to lower local redundancy. Because nearest-neighbor diversity is sensitive to source-set breadth, D is always interpreted jointly with Elo rather than as a stand-alone metric.

5.3 Token Observability

A third axis records the cost incurred to produce the quality and diversity signals so the tradeoff is auditable rather than implicit. Every generated idea and every judge call logs normalized input, output, and request counts; the five expenditure categories (graph construction, graph traversal, Elo judging, diversity judging, and rubric judging) are reported separately because their cost profiles differ qualitatively. The full normalization conventions and per-step totals are deferred to Appendix A (Table 8).

Set	Poster Elo	Oral Elo	Gap	Matches
ICLR 2025	967.0	1033.0	66.0	1000
ICLR 2026	982.5	1017.6	35.1	1000

Table 1: Oral-vs-poster calibration. The gap is oral minus poster mean Elo.

6 Results

6.1 The Judge Recovers a Known Quality Distinction

The Elo protocol reliably recovers the externally validated ICLR oral-vs-poster ranking in two independent calibration tournaments (Figure 2). ICLR 2025 oral ideas reach a mean Elo of 1033.0 versus 967.0 for posters, a 66.0 Elo gap; ICLR 2026 oral ideas reach 1017.6 versus 982.5 for posters, a 35.1 Elo gap. These two values do not claim to capture absolute scientific merit, but they do establish that the protocol is sensitive to a known venue-quality distinction, and they fix the interpretive scale for the generated-idea comparisons that follow.

6.2 Agentic Ideation Beats LLM-only Across All Settings; ReprNet and Abstracts Agents Tie on Quality

Both agentic arms improve mean Elo by $+56.8$ to $+123.0$ points over the shuffled LLM-only baseline across all three benchmarks (Table 2, Figure 3); the gap exceeds the calibrated oral-vs-poster reference (66.0 and 35.1 Elo) in every setting and surpasses the ICLR 2026 oral anchor in both random-ICLR tournaments. When the ReprNet and Abstracts agents are compared against each other in a single common-scale tournament per setting, however, the two arms are statistically indistinguishable on quality at every benchmark: $+5.4$ Elo at random10 (tournament rms $\approx$ 5.8), -0.8 Elo at random20 (rms $\approx$ 7), and -7.8 Elo at the Nature+ICLR mixed setting (rms ≈ 8.2). The entire agentic-vs-LLM-only advantage is therefore carried by the agentic loop itself—iterative tool use, deliberation, and self-critique over a 30–35-call budget—rather than by the representation graph.

The shape of the agentic gap is itself informative. In random10, the agentic advantage over LLM-only is smallest ($+56.8$ Elo); in random20 it is largest ($+123.0$); in the cross-domain mixed setting it sits in between ($+93.5$). This non-monotonic pattern is consistent with the agentic loop benefitting most when each idea has a moderately broad source context (random20's 20-paper pool) and proportionally less when the pool is narrower (random10) or harder (mixed); the ReprNet agent adds no further quality lift on top of this trend at the matched budget.

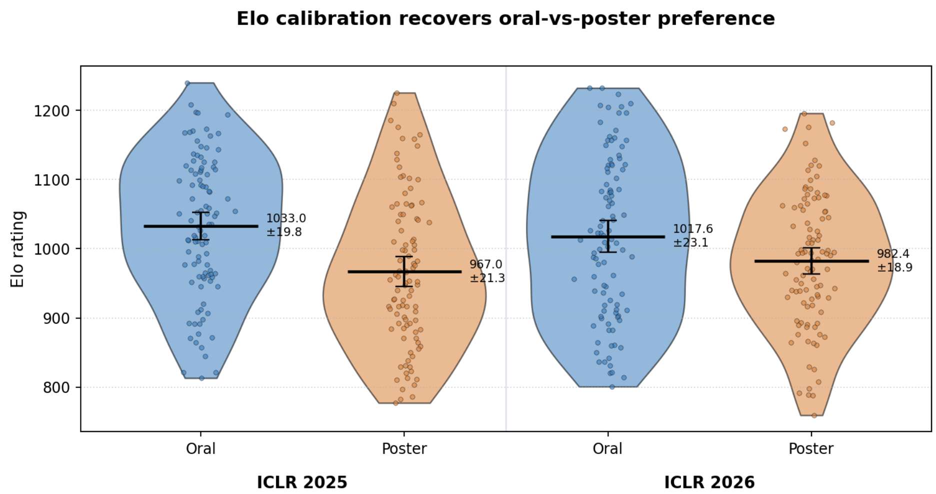

Figure 2: Calibration: the Elo judge separates ICLR oral abstracts from poster abstracts by 66.0 Elo on the 2025 split and 35.1 Elo on the 2026 split, recovering the externally known venue-quality ordering. Violins show per-idea Elo distributions; rules and error bars are means with $\pm 1.96 \cdot \mathrm{SEM}$ 95% confidence intervals. These two gaps anchor the interpretive reference scale for every subsequent generated-idea comparison.

Setting	LLM-only	ReprNet	Abstracts	Anchor	ReprNet−LLM-only	Abstr−LLM-only	ReprNet−Abstr
Random10 ICLR	963.11	1019.88	1014.46	1005.10	+56.77	+51.35	+5.42
Random20 ICLR	916.22	1039.22	1039.99	1004.57	+123.00	+123.77	−0.77
Nature+ICLR mixed	935.07	1028.59	1036.34	—	+93.52	+101.27	−7.75

Table 2: Quality results from the half-budget multi-way tournaments (random10: 4-way, 1200 matches; random20: 4-way, 2400 matches; mixed: 3-way, 1800 matches). All Elo values within a setting are on a single common scale because they are rated against each other in the same tournament. Anchor Elo is the mean Elo of 100 ICLR 2026 oral ideas when included. Each LLM-only and ReprNet-agent pool contains 90–200 ideas; Abstracts-agent pools contain 200/100/90 at random10/random20/mixed. Tournament rms convergence is 5.8/7/8.2 Elo respectively.

6.3 The ReprNet Substrate Wins Diversity, Largest Cross-Domain

Although the ReprNet and Abstracts agents are tied on quality, the ReprNet agent wins on local diversity in every setting (Table 3). At random10, ReprNet $D = 5.62 \pm 0.22$ vs Abstracts agent $D = 4.75 \pm 0.26$ (gap $+0.87$); at random20, 6.85 ± 0.16 vs 6.58 ± 0.18 (gap $+0.27$); at the Nature+ICLR mixed setting, 6.38 ± 0.20 vs 5.09 ± 0.26 (gap $+1.29$). The largest ReprNet diversity edge thus occurs in the cross-domain regime, where every idea must bridge an unfamiliar source pair; the smallest occurs at random20, where the 20-paper source context already gives both agents broad raw material. All three ReprNet-vs-Abstracts gaps exceed the combined 95% CIs, so the diversity edge survives similarity-judge noise.

The diversity advantage also distinguishes the ReprNet agent from a graph-free control of the agentic loop. LLM-only vs Abstracts agent gaps ($+0.47$, $+0.27$, $+0.72$ respectively at random10, random20, mixed) confirm that some of the diversity gain comes from the agentic loop alone—an iterating agent samples a wider idea space than a single LLM-only call—but the remaining ReprNet-over-Abstracts edge ($+0.87$, $+0.27$, $+1.29$) is larger still, especially cross-domain, and is the cleanest evidence in this paper for a genuine contribution of explicit graph structure beyond agency.

The interdisciplinary rubric corroborates the Elo and diversity results, locating the gain in integration rather than in surface domain coverage (Table 4). The ReprNet agent improves four of the five axes, with the largest movements in integration depth (4.23→4.41) and bidirectional value (3.93→4.12)—both of which target whether the domains genuinely shape each other, rather than whether both happen to be mentioned. Feasibility rises only marginally (2.64→2.73), so neither representation has yet solved the well-known feasibility gap. Source grounding decreases modestly (4.32→4.09), revealing an interpretable trade: graph-conditioned ideas synthesize more deeply across sources but attribute back to specific abstracts less explicitly than direct prompting.

6.4 Quality–Cost Tradeoff

The agentic quality gain comes at a 43–95× generation-token premium over the shuffled LLM-only baseline, concentrated in input-side traversal context rather than output length (Table 5; full per-step totals in Appendix A, Table 8). Per-idea generation tokens for the ReprNet agent are 282.8k, 380.4k,

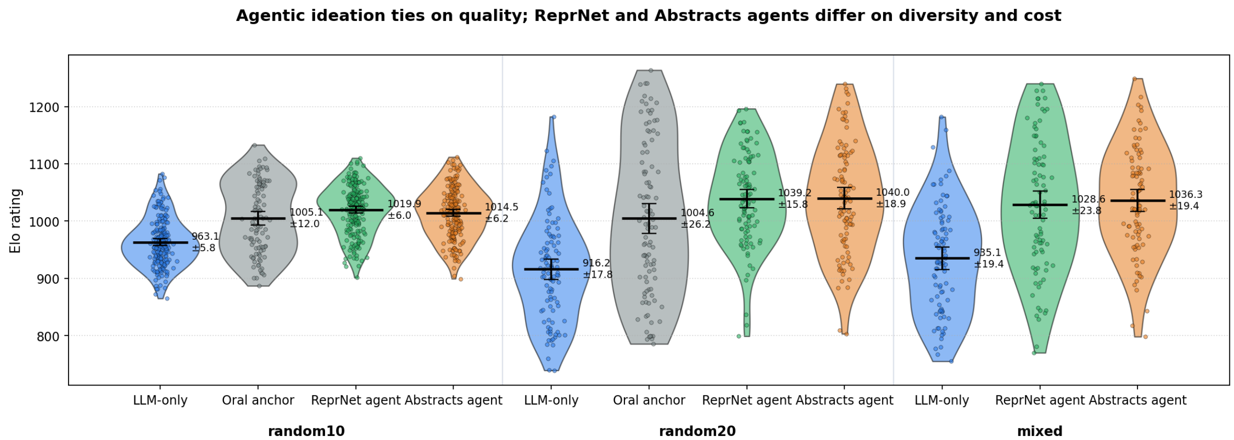

Figure 3: Per-idea Elo distributions by setting and arm. Both agentic arms (ReprNet and Abstracts agents) beat the shuffled LLM-only baseline by $+57$–$+123$ Elo and reach or exceed the ICLR 2026 oral anchor in the within-domain settings; the ReprNet and Abstracts agents are visually and statistically tied within each tournament's noise floor (rms 5.8–8.2 Elo). Each panel is one half-budget multi-way round-robin tournament (random10: 4-way / 1200 matches; random20: 4-way / 2400 matches; Nature+ICLR mixed: 3-way / 1800 matches; no oral anchor for the mixed groups). Numeric values match Table 2.

Setting	LLM-only D	Abstracts D	KGAgent D	ReprNet D
Random10 ICLR	4.28 ± 0.24	4.75 ± 0.26	5.09 ± 0.25	$\mathbf{5.62 \pm 0.22}$
Random20 ICLR	6.31 ± 0.21	6.58 ± 0.18	6.68 ± 0.19	$\mathbf{6.85 \pm 0.16}$
Nature+ICLR mixed	4.37 ± 0.29	5.09 ± 0.26	5.81 ± 0.27	$\mathbf{6.38 \pm 0.20}$

Table 3: Diversity score $D = 10-$ average nearest-neighbor LLM similarity; the highest score per setting is in bold. Intervals are 95% confidence intervals on the mean, computed as $\pm 1.96 \cdot \text{SEM}$ over the per-pair similarity ratings ($n = 240$ pairs per arm at random10/random20; $n = 289$–314 at the mixed setting). All three ReprNet-vs-Abstracts gaps ($+0.87$, $+0.27$, $+1.29$) clear the combined CI widths, so the ReprNet-agent diversity edge over the Abstracts agent is statistically real in every setting, with the largest effect in the cross-domain mixed regime. KGAgent (generic-knowledge-graph ablation) sits between the Abstracts agent and ReprNet in all three settings (5.09, 6.68, 5.81 vs ReprNet's 5.62, 6.85, 6.38), confirming that ReprNet's typed Repr/Transform schema adds a real $+0.17$ to $+0.57$ diversity gain on top of whatever generic structured graph would buy, with the largest effect in the cross-domain mixed regime. At random20 a finer-grained ablation (NR: typed Repr/Transform but no realization edges) drops diversity to 6.30 ± 0.21, below the generic KG's 6.68 and barely above the LLM-only baseline's 6.31, showing the realization hierarchy carries almost all of the diversity edge.

Rubric axis	LLM-only	ReprNet
Domain coverage	4.90	4.97
Integration depth	4.23	4.41
Bidirectional value	3.93	4.12
Source grounding	4.32	4.09
Feasibility	2.64	2.73

Table 4: Interdisciplinary rubric means on a 1–5 scale.

and 271.8k in random10, random20, and mixed, against 3.0k, 5.4k, and 6.3k for the LLM-only baseline. Disaggregating by call type shows that this multiplier comes almost entirely from accumulated tool-call history and graph-retrieval context, so the bottleneck is the size of the context the agent reasons over, not the length of the ideas it writes.

The Abstracts agent sharpens what kind of agentic context is cheap and what kind is expensive, now across all three benchmarks. Under the matched 50 tool-call / 70 request budget shared with the ReprNet agent, Abstracts agent consumes 476.1k, 490.5k, and 486.1k tokens per idea at random10, random20, and Nature+ICLR mixed respectively—in every case *more* than the ReprNet agent's 282.8k / 380.4k / 271.8k while reaching statistically identical mean Elo (Table 5). Per-call counts are nearly identical across the two agentic arms ($\sim$ 30–36 requests per idea), so the cost difference is not a count-of-calls difference; it is a payload difference. Each `read_paper`, `search_papers`, or `bridge` return carries one or more full paper abstracts back into the agent's context, whereas the graph tools (`get_node`, `find_paths`, etc.) typically return a handful of CamelCase node names and short descriptions, so the per-call payload is roughly an order of magnitude smaller on the graph substrate.

The cost advantage of the ReprNet agent is consistent and largest cross-domain. The Abstracts/ReprNet ratio is $1.68\times$ at random10, $1.29\times$ at random20, and $1.79\times$ at Nature+ICLR mixed—the same regime where the ReprNet agent also delivers its largest diversity edge. Across the three benchmarks combined, the ReprNet agent saves on the order of 130–200k tokens per idea at matched quality. In other words, the ReprNet agent's value under matched-quality comparison is not better mean Elo but (i) better local diversity, and (ii) the ability to deliberate over a compact symbolic substrate rather than re-reading raw abstracts every time the agent wants to look something up.

This per-idea comparison is a *generation-time* accounting and deliberately excludes the ReprNet graph's one-time, per-group construction, which is itself a substantial multi-turn LLM cost (reported separately in Table 8); the Abstracts agent's only comparable preprocessing is lightweight sentence embeddings. The headline "1.29–1.79× cheaper" claim should therefore be read at the per-idea generation margin, not as a total-cost-of-ownership statement: the upfront graph build dominates for small idea pools, and the net advantage materializes only once a group's graph is amortized over enough generated ideas. Quantifying that break-even point per setting is left to future work.

This cost structure positions two follow-on directions. First, traversal-context compression for the ReprNet agent—cached subgraph summaries, sparser retrieval, principled early stopping—remains the highest-leverage way to bring the absolute token premium down further. Second, for any non-graph agentic baseline, the natural symmetric move is retrieval over distilled paper representations (titles, keyword sets, or short autogenerated digests) rather than full abstracts, which would make the comparison cleaner on the cost axis. We did not run the distilled Abstracts-agent variant in this paper.

6.5 Ablation: Typed Schema vs Generic Knowledge Graph

The Abstracts agent comparison shows the agentic loop carries the quality gap over the LLM-only baseline, but leaves open whether ReprNet's residual diversity-and-cost edge comes from *any* structured graph at all or specifically from its Repr/Transform typed schema (Repr = noun, Transform = verb, realization edges for "is-a", typed input/output/composer edges). To isolate the schema contribution we run a third ablation across all three benchmarks: **KGAgent**, a matched agent over a *generic* knowledge graph built by the same LLM at matched cost. The generic KG replaces ReprNet's two node types with a single untyped Entity node and replaces typed edges with free-form string predicates (e.g. `is_a`, `uses`, `applies_to`, `outputs`). The KGAgent's tool palette mirrors ReprNet's read-only graph tools, and per-idea token cost is matched to the ReprNet ReprNet agent at every setting (random10: 283k vs 282.8k, random20: 299k vs 380k — KGAgent is actually *cheaper* here; mixed: 309k vs 272k tokens per idea, within 14%), so quality differences cannot be attributed to budget.

To keep the comparison clean, the KG itself is built by an **agentic** construction loop that mirrors ReprNet's per-paper graph-build pipeline (`repr_net.add_content`): the same LLM, a parallel multi-turn workflow, the same per-paper request budget (16), and a parallel tool palette (`search_entities`, `find_entities`, `get_entity`, `add_triple`). The only deliberate difference between the two builders is the schema. The resulting agentic-built KGs have 72.5 entities and 67 edges on average (range 68–76), versus ReprNet's ~ 62 nodes and 79–96 edges — a fair size match. We also report results for a non-agentic baseline KG built by a single-shot extractor; that variant has 86.8 entities on average because its lack of cross-paper search produces less entity reuse.

In head-to-head 2-way Elo tournaments against ReprNet, the agentic-built KGAgent loses by $+28.76$ to $+39.34$ Elo across the three settings (Table 6): random10 980.33 vs 1019.67 ($+39.34$, rms 7.20, $\approx 5.5\sigma$); random20 985.62 vs 1014.38 ($+28.76$, rms 10.47, $\approx 2.7\sigma$); Nature+ICLR mixed 981.27 vs 1018.73 ($+37.46$, rms 10.31, $\approx 3.6\sigma$). All three gaps clear the tournament noise floor. At random10 we also report results for a non-agentic one-shot-built KGAgent (987.04 vs 1012.96, $+25.92$, rms 7.05): the gap stays 25–40 Elo across both build variants, so the typed-schema effect is robust to KG construction methodology. Diversity differences mirror the Elo gaps, with ReprNet ahead by $+0.53$, $+0.17$, $+0.57$ at random10, random20, mixed (Table 3), and the ReprNet vs KGAgent CIs cleanly separated at random10 and mixed; the random20 gap is the smallest because the 20-paper context already gives both agents broad raw material.

Decomposing the schema: the realization hierarchy carries most of the effect. The KGAgent ablation collapses two schema mechanisms at once — typed Repr/Transform nodes *and* the realization (is-a) edge class — into a single generic-graph arm. To separate the two we run a third variant at random20, called **NR** (No Realization), that keeps the typed Repr/Transform nodes and their structural input / output / composer edges but removes the realization edge class entirely (builder cannot create abstraction edges, ideation agent cannot navigate parents). Construction is otherwise identical to the KGAgent fair builder, and per-idea cost is matched at 338k tokens vs ReprNet's 380k ($\approx 11\%$ cheaper). In a 1200-match Elo tournament NR loses to ReprNet by 990.09 vs 1009.91 ($+19.82$ Elo, rms 11.23, $\approx 1.8\sigma$); diversity drops to $D = 6.30 \pm 0.21$ — *below* even the KGAgent's 6.68 and barely above the LLM-only baseline's 6.31 (Table 7).

Comparing within the random20 setting, the Elo gap shrinks by $\approx 31\%$ when realization is removed (from KGAgent's $+28.76$ to NR's $+19.82$), so typed structural edges (composer / input / output) account for about a third of the schema's Elo edge. But diversity moves in the opposite direction: removing realization *hurts* diversity more than removing typing did (-0.55 from ReprNet vs -0.17 for KGAgent), pointing to the realization hierarchy as the source of ReprNet's diversity advantage. A plausible reading is that KGAgent's free-form predicates accidentally include weak abstraction-like relations (`is_a`, `applies_to`, `generalizes_to`) that the agent can use as fall-back abstraction navigation, whereas NR explicitly forbids any abstraction edge of any kind and the agent gets stuck in local structural neighbourhoods. Together, the two ablations decompose the schema effect into a structural-typing contribution (about a third of the Elo gap) and a much

Setting	LLM-only	ReprNet	Abstracts	ReprNet/LLM-only	Abstracts/ReprNet
Random10 ICLR	3.0k	282.8k	476.1k	95.1×	1.68×
Random20 ICLR	5.4k	380.4k	490.5k	71.0×	1.29×
Nature+ICLR mixed	6.3k	271.8k	486.1k	43.0×	1.79×

Table 5: Per-idea generation tokens by arm. Each entry includes input and output tokens for the full idea-generation call sequence (excluding one-time graph construction). Across all three benchmarks the Abstracts agent consumes 1.29–1.79× more tokens per idea than the ReprNet agent at statistically tied mean Elo; the gap is largest in the cross-domain mixed setting, where the ReprNet agent's symbolic substrate compresses paper content most aggressively.

Benchmark (build)	KGAgent Elo	ReprNet Elo	ΔElo	rms	$\approx \sigma$	ΔD	cost k tok (KG / Repr)
random10 (agentic)	980.33	1019.67	+39.34	7.20	5.5	+0.53	283.3 / 282.8
random10 (one-shot)	987.04	1012.96	+25.92	7.05	—	—	—
random20 (agentic)	985.62	1014.38	+28.76	10.47	2.7	+0.17	298.7 / 380.4
Nature+ICLR (agentic)	981.27	1018.73	+37.46	10.31	3.6	+0.57	308.6 / 272.0

Table 6: Typed schema vs generic knowledge graph, head-to-head 2-way Elo across all three benchmarks. ΔElo and ΔD are ReprNet minus KGAgent (positive favours the typed schema); rms is the per-arm tournament noise and $\approx \sigma$ the gap in noise units. Per-idea cost is matched within $\leq 22\%$ and at random20 the typed arm is the *more* expensive one, ruling out a budget confound. The one-shot-built KGAgent row shows the effect survives a change of construction methodology.

random20 arm	ΔElo	D	cost k tok
LLM-only baseline	—	6.31	5.4
KGAgent (generic, untyped)	+28.76	6.68	298.7
NR (typed, no realization)	+19.82	6.30	338.0
ReprNet (typed + realization)	ref	6.85	380.4

Table 7: Decomposing the schema at random20. ΔElo is the gap to ReprNet (positive = ReprNet ahead). Adding typed nodes back (KGAgent→NR) closes $\approx 31\%$ of the Elo gap; adding the realization hierarchy back (NR→ReprNet) closes the remaining $\approx 69\%$ and recovers essentially all of the diversity. NR's $D = 6.30$ falls *below* the untyped KGAgent, isolating the realization hierarchy as the source of ReprNet's diversity edge.

larger realization-hierarchy contribution (the rest of the Elo gap plus essentially all of the diversity gap).

The typed schema therefore does measurable work beyond what any matched-cost matched-build structured graph would buy, and the NR decomposition pins down which part of the schema does it. Three mechanisms in ReprNet are absent from the generic KG: **(i) Transform nodes give methods first-class identity** rather than burying them as predicates on noun-noun pairs; **(ii) typed** $(in\_nodes, out\_nodes, composer)$ **edges expose method composition explicitly**, allowing the agent to follow "method X composes method Y" without flattening into a predicate chain; **(iii) realization edges form a clean is-a hierarchy** distinct from other predicates, giving the agent a separate axis for abstraction. The NR ablation, which keeps (i)+(ii) but removes (iii), shows that the abstraction hierarchy of (iii) is the dominant mechanism: it carries roughly two thirds of the Elo gap and essentially all of the diversity gap. The typed structural edges of (i)+(ii) still contribute the remaining third of the Elo gap. The consistency of the schema effect across all three benchmarks — both within-domain and cross-domain, and robust to construction methodology — together with this internal decomposition is the cleanest evidence in this paper that the schema, not the *presence* of graph structure, is what gives ReprNet its residual edge.

7 Discussion

The agentic-vs-LLM-only advantage is consistent enough across benchmarks to rule out the usual confounders that complicate ideation results. The effect persists in three independently constructed benchmarks, holds in both within-domain (random ICLR) and cross-domain (Nature+ICLR) settings, and—unlike many ideation studies—survives a deliberately strengthened baseline in which the LLM-only arm sees the same source identities under deterministically reshuffled abstract order. Positional artifacts, source-pool drift, and differential exposure cannot explain the agentic gain.

The calibrated scale places this advantage in a regime where it is interpretable, not just statistically separated. LLM-as-judge protocols are noisy in absolute terms, yet the same protocol correctly separates ICLR orals from posters by 66.0 and 35.1 Elo points across two consecutive years; the agentic-vs-LLM-only gaps of 56.8–123.0 Elo therefore exceed the calibrated reference gap by 0.9–$3.5\times$. A margin of this size, together with concurrent diversity gains and deeper integration on the interdisciplinary rubric, supports reading the effect as a qualitatively different class of ideas rather than a marginal stylistic improvement.

The Abstracts agent answers the natural follow-up question cleanly. Across all three settings, the ReprNet and Abstracts agents are statistically indistinguishable on mean Elo (gaps of $+5.4$, -0.8, and -7.8 Elo, each within its tournament rms), so the entire agentic-vs-LLM-only gap is carried by the agentic loop itself rather than by the representation graph. This is a strong, falsifiable finding: ablating away the graph at matched compute does not move judged quality. At the same time, the graph contributes a real and consistent improvement on a different axis: nearest-neighbor diversity is $+0.27$ to $+1.29$ higher under graph conditioning than under Abstracts agent, with the largest effect in the cross-domain Nature+ICLR setting and the smallest in the within-domain random20 setting. The graph also delivers this advantage at lower cost: Abstracts agent burns 1.29–$1.79\times$ more tokens per idea than the ReprNet agent at matched quality.

The most useful way to interpret the cost–quality–diversity triangle is therefore as follows. The agentic loop is the right substrate for LLM scientific ideation; one-shot prompting is not. Within that substrate, choosing a symbolic graph representation over re-reading raw abstracts buys a consistent diversity bonus (largest where the agent has to bridge unfamiliar domains) and a consistent token discount (also largest cross-domain). These effects are smaller in absolute terms than the agentic-vs-LLM-only gap but more reliable across settings, and they identify the regimes where explicit graph structure is most worth investing in: cross-domain ideation and any setting where local idea-base redundancy matters. Conversely, in within-domain settings where the source pool is already broad (random20), the ReprNet agent buys little beyond cost efficiency, and a well-tuned non-graph agent may be a perfectly adequate substitute.

Cost compression is therefore not just about reducing the agentic arm's absolute token premium (43–95$\times$ over the LLM-only baseline). It is also about closing the Abstracts–ReprNet cost gap on the symbolic side, e.g. through cached subgraph summaries, sparser retrieval, and principled early stopping—and, symmetrically, on the flat side through retrieval over distilled paper representations rather than full abstracts. Both directions sit on the quality-equivalent Pareto frontier identified by this paper.

8 Limitations

Several limitations should temper the interpretation. **(i) LLM-judged quality.** All quality scores derive from LLM judges; exceeding the oral anchor inside an LLM-judged tournament should be read as a relative preference signal, not a claim that the generated ideas would be accepted as oral papers. Multi-judge robustness has not yet been confirmed. **(ii) Single generator.** All ideas are generated by the same underlying LLM; whether the agentic-loop dominance and the ReprNet diversity edge transfer to other model families is open. **(iii) Abstracts, not executed projects.** The benchmarks evaluate generated paper-style abstracts rather than experimentally validated research projects. **(iv) Sample size in the interdisciplinary setting.** 90 ideas per arm at mixed yields wider intervals than the within-domain random ICLR pools, even though the direction and ordering remain consistent. **(v) Persistent feasibility gap.** Feasibility on the rubric remains modest in both agentic arms (2.6–2.7 on a 1–5 scale), suggesting that neither representation has solved the long-standing feasibility weakness in LLM ideation. **(vi) Cost.** Even the cheaper agentic arm (graph) costs 43–95$\times$ more tokens per idea than direct prompting; production deployment would require substantial optimization before the approach is preferable in cost-sensitive scenarios. **(vii) Half-budget multi-way Elo.** Adding Abstracts agent doubled the number of arms per tournament without doubling matches, leaving tournament rms at 5.8–8.2 Elo; some Abstracts-vs-ReprNet gaps (e.g. $+5.4$ at random10) sit close to that noise floor. Larger tournaments would tighten the inference at the same qualitative ordering.

9 Reproducibility

All experimental artifacts can be re-audited from the public layout under `paper/` and `experiments/`. Figures are authored as HTML and matplotlib sources rendered by `paper/scripts/build_assets.py`; numeric tables are extracted from local experiment outputs into `paper/data/results.json`; and the matched random benchmark can be re-audited end-to-end with `experiments/2026-05-12_iclr2025_random_fair_benchmark_v2/audit_benchmark.py--strict`. The cached graph-arm manifest records source network names, duplicate names, and SHA-256 fingerprint verification for all 30 imported graph groups, so any reproduction can independently confirm that no protected graph was mutated during the experiments.

10 Conclusion

Letting El Agente Creativo operate iteratively over a fixed pool of source papers yields measurably better and more diverse research ideas than directly prompting the same LLM on the same papers, and the agentic gain over the shuffled LLM-only baseline ($+56.8$ to $+123.0$ Elo across three benchmarks) exceeds a calibrated oral-vs-poster quality scale by up to $3.5\times$, surpassing an ICLR 2026 oral anchor on both within-domain random tournaments. Comparing the ReprNet agent against a matched non-graph agentic baseline (Abstracts agent) at every setting splits this gain cleanly: the agentic loop carries *essentially all* of the quality lift over LLM-only (the ReprNet and Abstracts agents are statistically indistinguishable, within tournament noise, at random10, random20, and mixed—an absence of a detectable gap rather than proven equivalence), while explicit graph structure delivers two distinct, complementary wins on top—a consistent nearest-neighbor diversity edge of $+0.27$ to $+1.29$ over Abstracts agent (largest cross-domain), and a 1.29–$1.79\times$ lower per-idea *generation* token cost at matched quality (also largest cross-domain; this margin excludes the graph's one-time construction, against which the Abstracts agent's embedding overhead is negligible). We therefore view agentic ideation as the right substrate for LLM-based scientific discovery and explicit graph structure over the literature as a consistent diversity-and-cost win on top of it, most impactful in cross-domain settings. Traversal-context compression and human-evaluation robustness are the clearest next milestones toward practical deployment.

Acknowledgments

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Step	In	Out	Total	Req.
V2 LLM-only+ReprNet ideas (random10, random20)	94.63M	1.10M	95.72M	7891
V2 Elo (random10, random20)	6.78M	0.92M	7.70M	4000
V2 diversity (random10, random20)	2.05M	0.61M	2.66M	1932
Abstracts agent ideas (random10, n=200)	**94.51M**	**0.71M**	**95.22M**	**7081**
Abstracts agent ideas (random20, n=100)	**48.69M**	**0.36M**	**49.05M**	**3532**
Abstracts agent ideas (mixed, n=90)	**43.41M**	**0.33M**	**43.74M**	**3224**
Mixed augment	3.06M	0.11M	3.17M	407
Mixed LLM-only	0.51M	0.06M	0.57M	90
Mixed ReprNet	24.16M	0.30M	24.46M	2391
Mixed Elo	6.83M	0.88M	7.70M	3533
Mixed diversity	0.67M	0.20M	0.87M	600
Mixed rubric	0.12M	0.06M	0.18M	183

Table 8: Observed token usage by major step. The three Abstracts-agent rows cover 390 newly-generated ideas across the three benchmarks (200+100+90); their combined cost is 188M tokens versus a roughly estimated 110M for the ReprNet agent on the same idea counts, consistent with the per-idea Abstracts/ReprNet ratios reported in Table 5.

A Methodology Details and Reproducibility

This appendix collects implementation-level details deferred from the main text: the Bradley–Terry Elo update equations, the read-only graph duplication protocol, the token-observability conventions, and the per-step token-usage table.

A.1 Bradley–Terry Elo update

All ideas in a setting begin at Elo $R_i^{(0)} = 1000$. Each Swiss match contains $m = 4$ ideas; the LLM judge returns a complete ranking which is decomposed into all $m(m-1)/2$ ordered pairs. For a pair (i, j) with current ratings R_i and R_j, the Bradley–Terry expected score [Bradley and Terry, 1952] and the rating update given the pairwise outcome $S_i \in \{0, 1\}$ are

$$E_i = \frac{1}{1 + 10^{(R_j - R_i)/400}}, \qquad R_i \leftarrow R_i + \frac{K}{m-1}(S_i - E_i), \tag{2}$$

with $K = 32$. Mean final Elo per arm is reported with nonparametric bootstrap 95% confidence intervals over idea-level Elo values. The same protocol is first run on the ICLR oral-vs-poster calibration sets so every generated-idea gap can be read against a known-quality reference scale.

A.2 Read-only graph protocol

Prior experiment graphs are treated as read-only artifacts so any quality gain attributable to graph conditioning cannot be confounded by silent graph mutation. Each reuse of a cached graph proceeds in three steps. First, the source network is identified by name and a SHA-256 fingerprint is computed over the canonicalized node and edge payloads. Second, the source is duplicated into an experiment-scoped namespace (e.g. `reprgraph_fair_20260512_v2_*` for the V2 random ICLR benchmark, and `reprgraph_interdisciplinary_20260512_*_augmented` for the mixed benchmark's augmented copies); the duplicate's fingerprint is recomputed and required to match the source. Third, only the duplicate is exposed to write operations (Nature-abstract insertion in the mixed setting); the source remains untouched. Source name, duplicate name, and both fingerprints are logged in each experiment's `cached_graph_arm_manifest.json` so any reproduction can independently confirm that no protected graph was mutated.

A.3 Token observability

Every generated idea and every judge call records normalized usage fields `input_tokens`, `output_tokens`, `total_tokens`, and `requests`. OpenAI-compatible chat completions are read from `response.usage`; Pydantic-AI graph-construction runs are read from the native `RunResult.usage()` object. Per-idea usage is stored on each idea JSONL row, and aggregate graph-build usage is additionally attached to MongoDB network metadata. Five expenditure categories (graph construction, graph traversal, Elo judging, diversity judging, and rubric judging) are tracked separately because their cost profiles differ qualitatively; their per-step totals across the paper's experiments are reported in Table 8.

A.4 Generic KG Ablation: Schema, Builder, Ideation, and Fairness

This subsection collects the implementation details of the generic-KG ablation reported in §6 and discusses why each element is matched to the ReprNet agent. The goal of the ablation is to isolate a single variable — the schema of the substrate that the ideation agent reasons over — so every other degree of freedom must be controlled. Three components define the arm: the generic KG schema, the agentic per-paper KG construction loop, and the KGAgent ideation agent.

Generic KG schema. ReprNet's substrate uses two node types (Repr nouns and Transform verbs), a typed (in_nodes, out_nodes, composer) tuple on each Transform, and a dedicated `realization` edge class for "is-a" relations. The generic KG removes all of these distinctions. It has a single untyped `Entity` node — any concept (data type, method, dataset, metric, etc.) is an Entity — and a single edge class carrying a free-form `snake_case` predicate string (e.g. `is_a`, `uses`, `applies_to`, `outputs`, `is_evaluated_on`, `improves_upon`, `trained_on`). There is no fixed predicate vocabulary; the builder LLM chooses whichever string it judges most apt for each triple. This is the standard generic knowledge-graph representation and is what an off-the-shelf LLM would produce if asked to "extract a knowledge graph" from a paper. Crucially, *method composition collapses*: a Transform node with ≥ 2 inputs, ≥ 1 output, and a composer in ReprNet becomes a chain of ≥ 3 generic predicates on noun-noun pairs, losing first-class identity. The hierarchy collapses too: `realization` as a dedicated edge type becomes `is_a` as one predicate among many, which the agent must disambiguate from `uses`, `applies_to`, etc. This collapse is the single variable under test.

Agentic construction loop. ReprNet's KG builder (`repr_net.add_content`) processes one paper at a time through a Pydantic AI agent that issues multiple tool calls per paper, with a hard per-paper request budget of 16 and a system prompt that instructs the LLM to search first, reuse exact node names where possible, and add only the central new nodes. The fair generic-KG builder (`kg_builder_agentic.py` in the ablation experiment directories) mirrors this loop step-for-step: the same generator LLM (`aws/anthropic/bedrock-claude-sonnet-4-6`); the same Pydantic AI driver with `UsageLimits(request_limit=16)`; the same workflow shape in the system prompt (read the abstract, search first, decide which existing entities to reference, add 5–7 triples per paper, then verify); and the same cross-paper state object that accumulates entities and triples across the group. The only documented difference between the two builders is which schema they emit. We additionally report results for a one-shot baseline builder (single LLM call per paper returning 5–7 triples, no cross-paper search) to expose how much of the ablation outcome depends on the construction methodology; §6 shows the typed-vs-generic gap is robust across both build modes.

Builder tool palette. The fair builder exposes five tools, each a direct counterpart of a ReprNet builder tool: `search_entities(query, top_k)` (embedding-based; mirrors ReprNet's `search_similar_nodes`), `find_entities(pattern)` (substring; mirrors `find_nodes`), `get_entity(name)` (inspect node + adjacent triples; mirrors `get_node`), `add_triple(subject, predicate, object)` (mirrors ReprNet's `add_repr`/`add_transform`/`add_realization` combined, since generic edges carry no type), and `list_recent_additions` (book-keeping helper). The search tool uses the same embedding provider (`text-embedding-ada-002` via NVIDIA's inference endpoint) that the rest of the pipeline uses for similarity scoring.

Resulting KG sizes. At random10 the fair-built generic KGs average 72.5 entities and 67 edges per group (range 68–76), versus ReprNet's ~ 62 nodes and 79–96 edges — a close size match. The one-shot baseline produces larger but more redundant graphs (86.8 entities on average) because its lack of cross-paper search reduces entity reuse; this size gap is exactly what the agentic builder fixes. At random20 and the mixed setting, the same per-paper budgets produce graphs in the same per-paper-density regime; we deliberately do not re-tune the budget across settings because the ReprNet builder does not re-tune either.

KGAgent ideation agent. The downstream ideation agent (`kg_agent.py`) is constructed by direct parallel to the ReprNet agent. It uses the same generator LLM, the same Pydantic AI framework, the same `IdeaProposal` Pydantic output schema (single title + abstract field), the same 50-tool-call / 70-request ideation budget (constants `TOOL_CALL_LIMIT` and `REQUEST_LIMIT` in the experiment driver), and the same workflow shape in the methodology prompt: pick random entities, inspect, explore neighbours, find paths, optionally search at another abstraction level, synthesize, self-critique. The tool palette is the read-only counterpart of the builder palette: `random_entity(min_degree)`, `get_entity(name)` (full view: predicates, in/out neighbours, source papers), `search_entities(query, top_k)`, `get_neighbors(name)`, and `find_paths(src, dst)`. Each of these maps onto a ReprNet read-only tool with matched semantics, so the agent's reasoning loop has the same shape on both substrates. The mapping is not strictly one-to-one in cardinality, however: the ReprNet read-only palette is somewhat richer (around ten tools), exposing separate parent/child hierarchy navigation (`get_parents`/`get_children`), component and wide-node analysis, and a second path tool (`find_shortest_path`) in addition to the five above. Most of these extra affordances exist precisely because the typed schema supports them—hierarchy navigation requires the realization edges that the generic KG lacks—so they are best read as part of the substrate under test rather than a neutral tooling advantage; we nonetheless flag that the ReprNet arm exposes more tools, so we cannot fully exclude a tool-count contribution to its edge.

Degree of freedom	ReprNet	KGAgent (fair build)
Source paper IDs per group	V2 random / mixed	verbatim copy
Generator LLM (builder + ideator)	Claude Sonnet 4.6 (Bedrock)	same
Pydantic AI framework + output schema	`IdeaProposal`	same
Ideation tool budget	50 tool calls, 70 req	same constants
Ideation workflow shape	random→inspect→explore→bridge→synth→critique	identical
Ideation tool palette (read-only)	random / get / search / neighbours / paths (+ hierarchy/analysis tools)	matched core; ReprNet richer
Per-paper KG-build LLM + driver	Pydantic AI multi-turn	same
Per-paper KG-build request budget	16 requests	same (`request_limit=16`)
Builder tool palette	search / find / get / add (typed)	search / find / get / add (untyped)
Per-paper token budget	natural outcome of above	matched within $\leq 14\%$
Substrate schema (variable under test)	**Repr+Transform typed**	**Entity + free predicate**

Table 9: Fairness gates for the generic-KG ablation. Every degree of freedom is matched between the two arms except the substrate schema, which is the variable under test. The "KGAgent (fair build)" column refers to the agentic-built generic KG used at every reported setting; §6 additionally reports a one-shot-built variant at random10 to verify that the typed-vs-generic effect is not an artifact of how the KG was assembled.

For the mixed benchmark the methodology prompt is the interdisciplinary variant (must use one ML and one Nature-domain concept; explain how the scientific mechanism shapes the ML representation), again matched to the ReprNet agent.

Fairness gates. Table 9 enumerates every degree of freedom of the comparison and marks which arm controls it. All but one — the schema — are matched by construction; the schema is the deliberate variable under test. Per-idea token cost is the most easily-confounding variable and is reported in three places: random10 (283.3k for fair KG vs 282.8k for ReprNet, within 0.2%), random20 (298.7k vs 380.4k, KG is 22% *cheaper*), and mixed (308.6k vs 272k, 14% more expensive). In every case ReprNet's quality edge cannot be attributed to spending more compute per idea: at random20 the more expensive arm is the typed-schema one, and yet KGAgent still loses by $+28.76$ Elo. This rules out the standard "maybe the better arm just had more budget" confound.

Why each gate matters. Each row of Table 9 closes a specific alternative explanation for the observed gap. Matching the source paper IDs rules out content-pool differences. Matching the generator LLM rules out model-capability differences (e.g. one arm using a stronger model). Matching the Pydantic AI framework and `IdeaProposal` schema rules out output-format effects (JSON-validity rates, field-length nudges). Matching the ideation budget rules out "more tool calls → better ideas". Matching the workflow shape rules out prompt-engineering luck — the same inspect/explore/bridge/synthesize/critique scaffold is on both sides. Matching the core tool semantics limits the "ReprNet just exposes more affordances" confound, though it does not eliminate it: both agents share the same five-verb reasoning loop (random/get/search/neighbours/paths) with matched semantics, but the ReprNet read-only palette is somewhat larger (around ten tools, adding explicit parent/child hierarchy navigation, component/wide-node analysis, and a shortest-path tool). Those extra tools are schema-specific—they operate on the realization hierarchy and typed structure that the generic KG does not have—so they are part of the substrate under test; still, we do not claim strict numeric tool parity, and a residual affordance advantage for ReprNet cannot be ruled out. Matching the construction loop (multi-turn agent with `request_limit=16` and search-first instructions) rules out the construction-quality confound that motivated this ablation in the first place — without the agentic builder, the comparison would be against a weaker one-shot generic KG and any quality gap could be blamed on construction. Matching the per-paper token budget closes the same loophole at the cost level.

The single remaining systematic difference is therefore the schema, and the empirical effect of that single difference (Table 3 and §6) is a $+28.76$ to $+39.34$ Elo gap and a $+0.17$ to $+0.57$ diversity gap consistent across all three benchmarks. Three concrete schema mechanisms are absent from the generic KG and plausibly account for the gap: Transform nodes give methods first-class identity rather than burying them as predicates; (in_nodes, out_nodes, composer) edges expose method composition as a typed structural pattern rather than a flattened predicate chain; and a dedicated `realization` edge class gives the agent a clean abstraction axis distinct from other predicates. We discuss which of the three carries which fraction of the gap as a follow-on in §6.

A.5 Decomposition Ablation: Typed Schema Without Realization (NR)

The KGAgent ablation removes two ReprNet mechanisms at once: the typed Repr/Transform node distinction *and* the realization (is-a) edge class. To attribute the schema effect to one mechanism or the other, we run an intermediate variant at random20,

Setting	Elo	Title	Idea sketch
Nature+ICLR mixed	1437.2	GFlowNet-Guided Cis-Regulatory Sequence Design from Plant MPRA Fitness Landscapes	Uses GFlowNets to design plant cis-regulatory sequences under massively parallel reporter assay feedback. The proposal targets epistatic promoter design and explicitly frames diversity preservation as a way to avoid active-learning mode collapse.
Nature+ICLR mixed	1399.7	Rotation-Equivariant Graph Networks for Predicting Chiral Phonon Angular Momentum Transfer in Anharmonic Crystals	Connects rotation-equivariant graph neural networks with chiral phonon angular-momentum conservation. The idea proposes predicting anharmonic phonon scattering and spin-lattice transfer more cheaply than first-principles perturbation calculations.
Random10 ICLR	1237.4	Memorization Asymmetry as a Privacy Risk Signal in Speculative Decoding	Treats the mismatch between draft-model and target-model memorization profiles in speculative decoding as a privacy signal. The proposed test measures whether accepted and rejected token proposals reveal memorized training examples.
Random10 ICLR	1229.1	In-Context Simulation-Based Inference via Autoregressive Score Matching on Markovian Trajectories	Recasts causal Transformers as amortized simulation-based inference engines over Markovian trajectories. The method would learn posterior estimation in context, reducing retraining when simulators or observations change.
Random20 ICLR	1184.4	Causally-Structured Sparse Attention for Doubly Robust Estimation from Continuous-Time Clinical Data	Builds sparse attention masks around temporal causal structure for treatment-effect estimation from ICU-style sensor streams. The proposal links transformer nuisance estimation with g-computation and inverse-probability weighting constraints.
Random20 ICLR	1182.0	Mechanistic Reverse-Engineering of Selective State Space Models on Algorithmic Tasks	Extends mechanistic interpretability from transformer circuits to selective state-space models such as Mamba. The idea asks whether SSM solutions to algorithmic tasks can be reverse-engineered into human-readable mechanisms and formal accuracy bounds.

Table 10: High-Elo examples from generated graph-conditioned ideas. Elo values are final idea-level ratings from the corresponding tournament outputs.

NR (No Realization), which keeps the typed Repr/Transform node types and their structural input / output / composer edges but *forbids any realization edge*. The builder has `add_repr` and `add_transform` tools (no `parent_names` parameter), but no `add_parent` / `add_realization` tool of any kind; the ideation agent has no parent/child navigation tool. All other elements — generator LLM, Pydantic AI driver, output schema, multi-turn construction with `request_limit=16`, ideation budget (50 tool calls, 70 requests), workflow scaffold — match the KGAgent fair builder and ideation agent one-to-one. The only systematic differences from ReprNet are (a) the absence of realization edges in the substrate and (b) the absence of any abstraction-navigation tool in the ideation agent.

The resulting NR networks average 114.6 nodes per random20 group (75.9 Repr + 38.7 Transform) and 142.8 structural edges (input 860, output 472, composer 96, summed across all ten groups), with 0 realization edges by construction. All $200/200$ papers are processed cleanly (no `request_limit` hits). Per-idea cost is 338k tokens, $\approx 11\%$ cheaper than the ReprNet agent's random20 cost (380k) and $\approx 13\%$ more expensive than the random20 KGAgent (299k).

In the head-to-head Elo tournament against ReprNet random20 (1200 matches), NR loses by 990.09 vs 1009.91 ($+19.82$ Elo gap, rms 11.23, $\approx 1.8\sigma$). Top-5% pairwise diversity yields $D = 6.30 \pm 0.21$ ($n = 240$ pairs). These numbers, set against the KGAgent and full-ReprNet arms, are collected in Table 7.

The headline interpretation is that the gap from KGAgent to NR (closing $\approx 31\%$ of the Elo gap by adding typing back) is attributable to typed structural edges, while the gap from NR to ReprNet (closing the remaining $\approx 69\%$ of the Elo gap and $+0.55$ in diversity) is attributable to the realization hierarchy. In particular, removing realization *below* the generic-KG baseline on diversity ($6.30 < 6.68$) is the key evidence: without realization the agent cannot find any abstraction-level bridges, and NR is the only ablation in which the agent has *strictly less* ability to navigate abstractions than even the free-predicate generic KG (whose predicate vocabulary accidentally contains abstraction-like relations such as `is_a`, `applies_to`, `generalizes_to`).

B High-Elo Generated Idea Examples

Table 10 gives representative high-Elo generated ideas from the ReprNet agent in each benchmark. The examples are selected from the final Elo JSONL files and exclude the ICLR oral-anchor calibration rows.

C Prompt Templates

This appendix records the natural-language prompt templates used by the LLM steps in the experiments. Runtime fields such as source abstracts, prior titles, graph-tool results, and idea text are shown as bracketed placeholders. Deterministic steps such as source sampling, embedding calls, graph fingerprinting, and aggregation do not use natural-language prompts.

C.1 Random ICLR LLM-only Baseline

```
System:
You are a research scientist generating new ML research ideas.
Propose one focused, feasible, high-impact idea for a small team.
Do not claim completed results. Do not copy an input paper.
Return valid JSON only.

User:
Method: [method]

[method note: random abstracts, embedding-diverse abstracts, or
embedding-similar abstracts; avoid merely averaging papers.]

Prior generated titles to avoid:
[last prior titles]

Source abstracts:
[paper title, keywords, and abstract blocks]

Return exactly:
{
  "title": "<short title under 20 words>",
  "abstract": "<150-250 word forward-looking paper-style abstract>"
}
```

C.2 Nature+ICLR LLM-only Baseline

```
System:
You are a senior interdisciplinary AI researcher. Generate one focused,
feasible research idea that genuinely integrates machine learning with
a scientific mechanism, measurement, constraint, or data process from
the supplied Nature abstracts. Return valid JSON only.

User:
Method: [mixed group prompt method]

Prior generated titles to avoid:
[last prior titles]

Source abstracts:
[10 ICLR abstracts and 5 Nature abstracts in deterministic per-idea
shuffled order]

Rules:
- Use at least one ICLR source and at least one Nature source.
- Do not merely propose applying a generic model to a scientific dataset.
- The Nature-domain content must change the ML representation, objective,
  evaluation, theory, simulator, or training signal.
- Be specific enough for a small research team to start implementation.

Return exactly this JSON object:
{
  "title": "<short title under 20 words>",
  "abstract": "<180-260 word forward-looking paper-style abstract>",
  "cross_domain_connection": "<one sentence explaining the concrete bridge>",
  "source_ids_used": ["<ids used>"],
  "risks": "<main feasibility or evaluation risk>"
}
```

C.3 Representation Graph Construction

```
System:
You are an expert at building representation networks. A representation network
models a scientific domain as a directed graph of Repr nodes
(representations / data types) and Transform nodes (processes that convert
representations).
```

```
Node types:
- Repr: a data type, concept, or artifact (noun), with name, optional
  description, refs, and parent names.
- Transform: a process that converts one or more Reprs into others (verb),
  with name, in_node_names, out_node_names, composer_names, optional
  description, refs, and parent names.

Workflow:
1. Read the abstract carefully to identify the paper's key concepts.
2. Search first with search_similar_nodes using several queries.
3. Reuse existing nodes when possible and add only needed new Repr and
   Transform nodes.
4. Add Repr nodes, then Transform nodes, then parent links.

Rules:
- Repr = noun; Transform = verb.
- Every Transform must have in_node_names, out_node_names, and composer_names.
- Reuse exact existing node names instead of creating duplicates.
- Set refs=[url] on the most central new node.
- Use CamelCase names.

Budgeted pilot constraints:
- For each paper, add only the core contribution.
- Add 2-4 central Repr nodes and 1-2 central Transform nodes.
- Prefer reusable abstractions over paper-specific duplicates.

User:
Add representations from the following paper to the network.

URL: [paper URL or source ID]
Abstract / Content:
[paper abstract]

Instructions:
1. Analyze the abstract to identify key concepts and contributions.
2. Use search_similar_nodes with multiple queries.
3. Add new Repr nodes first, then Transform nodes.
4. Use find_nodes to verify the new nodes exist.
```

C.4 Graph-Conditioned Ideation

```
System / methodology:
You are a creative interdisciplinary researcher generating ideas from a
representation graph that combines ICLR 2025 machine-learning papers with recent
Nature papers.

Your idea must:
- Use at least one machine-learning concept and at least one Nature-domain
  concept.
- Mechanistically integrate the domains; do not merely apply a generic model
  to a scientific dataset.
- Explain how the scientific mechanism, measurement, data process, or
  constraint changes the ML representation, objective, evaluation, simulator,
  or theory.
- Be feasible for a small research team in 6--12 months.
- Prefer ideas that can be evaluated using available scientific data,
  simulations, or controlled benchmarks.

Fixed workflow appended by the agent runner:
1. Pick random nodes using get_random_node.
2. Inspect the most interesting pair with get_node.
3. Explore parents and children briefly.
4. Find paths between nodes using find_paths or find_shortest_path.
5. Optionally search at another abstraction level.
6. Synthesize one research idea that bridges concrete nodes through a
   surprising connection.
7. Self-critique for feasibility, focus, reviewability, and natural language.

Task:
Generate a novel, feasible and impactful research idea by exploring the graph
of representations. Pick random nodes, explore hierarchy, find paths for
inspiration, then synthesize and self-critique before returning the final idea.

Output:
title: short descriptive title under 20 words.
abstract: one paragraph, 150-300 words, with problem, approach, specific
algorithms/datasets/metrics, expected contribution, and impact. Do not include
tool traces or internal graph node names.
```

C.5 Elo Preference Judge

```
System:
You are an expert research evaluator with deep knowledge of machine learning,
chemistry, and materials science. You will be given several research ideas and
must rank them from most interesting and promising to least. Prioritise ideas
that are genuinely novel, scientifically exciting, concretely actionable by a
small team in 6-12 months, and high-impact. Respond with valid JSON only.

Do NOT reward number of cited papers, author-year references, length or polish
of the abstract, or incremental extensions without a fresh angle.

User:
Rank the following [n] research ideas from most promising (rank 1) to least
promising (rank [n]).

Idea 1:
[title and abstract, truncated at 2000 characters]

...

Output ONLY valid JSON:
{
  "ranking": [<list of idea numbers from best to worst>],
  "reasoning": "<2-3 sentences>"
}
```

C.6 Diversity Similarity Judge

```
System:
You are an expert research analyst. Given two research ideas, rate how similar
they are in terms of core topic, methodology, problem formulation, and
potential applications. A score of 10 means the ideas are essentially the same
concept with minor wording differences. A score of 1 means they are completely
unrelated. Respond with valid JSON only.

User:
Rate the similarity between these two research ideas.

Idea A:
[title and abstract, truncated at 2000 characters]

Idea B:
[title and abstract, truncated at 2000 characters]

Rate their similarity from 1 (completely unrelated) to 10 (essentially
identical). Consider: core problem, methodology, target domain, and potential
applications.

Output ONLY valid JSON:
{
  "similarity_score": <1-10 integer>,
  "overlap_areas": "<brief description of what the ideas share>",
  "difference_areas": "<brief description of key differences>",
  "reasoning": "<2-3 sentence explanation of the similarity rating>"
}
```

C.7 Interdisciplinarity Rubric Judge

```
System:
You are a strict research-program evaluator. Score interdisciplinary
AI+science ideas using only the provided title and abstract. Return valid
JSON only.

User:
Idea title: [title]

Idea abstract:
[abstract]

Score each field from 1 (weak) to 5 (excellent):
- domain_coverage: substantively uses both AI/ML and Nature-domain science.
- integration_depth: domains interact mechanistically rather than superficially.
- bidirectional_value: contributes to ML methodology and the scientific domain.
- source_grounding: appears grounded in supplied paper mechanisms, not generic.
- feasibility: executable by a small team in 6--12 months.
```

```
Return JSON:
{
  "domain_coverage": <1-5>,
  "integration_depth": <1-5>,
  "bidirectional_value": <1-5>,
  "source_grounding": <1-5>,
  "feasibility": <1-5>,
  "rationale": "<brief explanation>"
}
```

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